Power Calculations

$V_{peak} = \sqrt{2} \cdot 120 = 169.7\ V$

$I_{inrush(no\ resistor)} = \frac{169.7}{\text{assume } 1\Omega} = 169.7\ A$

$I_{diode\text{-}surge\text{-}max} = 30\ A$

NTC size

t(0) resistance needed

$\frac{\text{Peak } V}{I_{diode\text{-}surge\text{-}max}} \rightarrow \frac{169.7}{30A} = 5.67\Omega$

Stay below max $\frac{169}{28A} = 6.06\Omega$

Steady-state needed

0.0546 A

Power draw

Pre amp: $170 \cdot .0132 = 2.245W$

Power amp: $170 \cdot .0285 = 4.845W$

$2.295 + 4.845W = 7.14W$

$I_s = \frac{7.14}{170V} = 0.042A$

$\times 1.30\ \%$

0.0546 A

ICL: B57211P0809N301

R@25°C : $8\Omega \pm 20\%$

Worst case low

$\frac{169.7}{6.4} = 26.5A$

Worst case high

$\frac{169.7}{9.6} = 17.68A$ ✓

$V_{ripple} = \frac{I}{2cf}$

$\frac{.0546}{2 \cdot 940\mu \cdot f} = 0.484V$

Steady-state current max: 4A ✓

R@current: $159m\Omega$

Vdrop@current

$.159\Omega \cdot (I_s + 30\%) = 0.009V$

0.0546A

$V_s = 169.7 - 0.009 = 169.69V$ ✓

SL 1522102

R@25°c : 220 Ω ± 25%

Low worst

$\frac{169.7}{165} = 1.02A$

high Worst

$\frac{169.7}{275} = 0.62A$

R steady-state : 2A ✓

V drop steady

$.990\,\Omega \cdot (.0546) = 0.054V$

$169.7 - .054 = 169.65V$